

Operating Systems

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SEC-5

WEEK-1

1. Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID (PID) of both parent and child processes.

Using Linux terminal commands (uname, lscpu, lsblk, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

Code

```
prudvi@LAPTOP-TEKO376B: ~ × +
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
#include <stdlib.h>

int main()
{
    char command[50];

    printf("Enter command: ");
    scanf("%s", command);
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int source, destination;
    char buffer[100];
    int n;

    source = open("source.txt", O_RDONLY);

    if (source < 0)
    {
        printf("Cannot open source file\n");
        return 1;
    }

    destination = open("destination.txt",
        O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (destination < 0)
    {
        printf("Cannot create destination file\n");
        close(source);
        return 1;
    }

    while ((n = read(source, buffer, sizeof(buffer))) > 0)
    {
        write(destination, buffer, n);
    }

    close(source);
    close(destination);

    printf("File copied successfully\n");

    return 0;
}
```

output

```
prudvi@LAPTOP-TEKO376B:~$ nano linux.c
prudvi@LAPTOP-TEKO376B:~$ gcc linux.c
prudvi@LAPTOP-TEKO376B:~$ ./a.out
Enter command: ls
Parent PID: 523
Child PID: 530
Process1.c floder1.c floder6.c folder1.c folder3 folder4.c folder6.c process.c process1.c
a.out floder3 folder folder2 folder3.c folder5 linux.c process.c.save process1.c.save
floder3 floder3.c folder1 folder2.c folder4 folder5.c peocess1.c process1 w1.c
Child completed
```